

Kola Nikhil Jai

Hyderabad, Telangana, India

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Experience

CDPL | *Software Engineering Intern*

Feb 2026 - Present

- Led development and production deployment of a scalable web platform on AWS, supporting 1K+ users and improving system performance by 35% through optimized backend services and efficient frontend rendering.
- Architected and maintained cloud infrastructure (AWS EC2, S3, autoscaling), achieving 99.9% uptime and enabling seamless handling of 2× traffic spikes.
- Contributed to building a defence-oriented drone simulation system, developing a VR-based immersive training environment integrated with controller inputs for real-time flight simulation and mission scenario testing.
- Engineered real-time control and physics interaction modules for the drone simulator, improving simulation accuracy and control responsiveness by 40%.
- Designed and developed interactive kiosk systems for user engagement and demonstrations, optimizing UI/UX workflows and improving interaction efficiency by 30%.
- Automated CI/CD pipelines and deployment processes, reducing release cycle time by 50%, and resolved 20+ system-level bugs, improving overall reliability by 25%.

Personal Projects

Respiratory Sound Diagnostic Engine (CNN-based)

Dec 2025

- Engineered a DSP pipeline using Librosa to convert raw respiratory audio into MFCC-based spectral features (6K+ dimensions), enabling robust feature extraction for medical classification.
- Designed and trained a multi-layer CNN (TensorFlow/Keras) with Conv2D, pooling, and dense layers, achieving accurate classification across 8 respiratory disease classes.
- Optimized training using Adam optimizer + normalization techniques, improving convergence stability and model generalization over 50 epochs.
- Integrated audio datasets with patient metadata (Pandas), enabling enriched feature learning and improved prediction context.
- Developed an end-to-end inference system with Tkinter GUI, supporting real-time audio input, prediction, and visualization of performance metrics.
- Implemented efficient model serialization and data pipelines, reducing inference latency and improving deployment readiness.

Moodly: Emotion-Aware Music Recommendation System | Demo

Sep 2025

- Engineered an end-to-end emotion-aware recommendation pipeline using DeepFace, OpenCV, and Python, converting real-time webcam input into mood-based music suggestions.
- Integrated facial emotion recognition with DeepFace + OpenCV (cv2), enabling accurate classification across multiple emotional states.
- Designed a scalable backend using Flask and Flask-CORS, exposing REST APIs for seamless frontend-backend communication.
- Developed a mood-to-music mapping engine and integrated Google YouTube Data API (googleapiclient) for real-time content retrieval.
- Built a responsive frontend using React.js, enabling real-time image capture and dynamic recommendation display.
- Implemented modular full-stack architecture with REST APIs, CORS, and API integrations, ensuring reliability and scalability.

Education

Vignan Institute Of Technology & Science

July 2024 - Present

Bachelor of Technology in Information Technology

Hyderabad, Telangana

Technical Skills

Languages: Python, Java, TypeScript, JavaScript

AI Systems: LLM Systems, Agentic AI, LangChain, LangGraph, Prompt Engineering, RAG Pipelines, Model Evaluation

Cloud & Infra: AWS (EC2, S3, Lambda), Serverless Architecture, CI/CD Pipelines, Docker

Distributed Systems: Kafka, Redis, WebSockets

Databases: PostgreSQL, MongoDB

Achievements / Certifications

- Scalar Certified: Java Mastering the fundamentals
- Winner – Hack With AI Hackathon (KL University)